

# Task 1: Titanic Survival Prediction

## Project Overview

This project predicts whether a passenger survived the Titanic disaster using a machine learning classification model. It analyzes passenger information, performs data preprocessing, trains the model, and evaluates its performance through predictions and visualizations.

## Objective

Build a machine learning model to accurately predict passenger survival based on historical passenger data.

## Technologies Used

- Python
- Pandas
- Matplotlib
- Scikit-learn
- Google Colab

## Workflow

- Import the required libraries.
- Load the Titanic dataset.
- Preprocess the data by handling missing values and encoding categorical variables.
- Split the dataset into training and testing sets.
- Train the Random Forest classification model.
- Evaluate the model using accuracy.
- Visualize the prediction results.
- Predict the survival status of a sample passenger.

## Result

The model successfully predicts passenger survival with good accuracy. The project also provides visualizations, including actual vs. predicted survival and survival distribution, to evaluate the model's performance.